“Titanic Dataset EDA”

A PROJECT SUBMITTED TO

SYNTECXHUB Technologies Kanpur Nagar, Uttar Pradesh

FOR THE INTERNSHIP

IN

DATA SCIENCE

By

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❖ INTRODUCTION

The Titanic Survival Analysis using EDA project is based on the historical data of passengers aboard the RMS Titanic sinking, which occurred in April 1912. The Titanic was one of the largest and most luxurious ships of its time, but it tragically sank after colliding with an iceberg during its maiden voyage. Due to the lack of sufficient lifeboats and emergency preparedness, a large number of passengers lost their lives, making it one of the most famous disasters in history.

The dataset used in this project contains detailed information about passengers, including their age, gender, ticket class, fare, family members on board, and survival status. This data provides an excellent opportunity to analyze real-world patterns and understand how different factors influenced survival chances.

Exploratory Data Analysis (EDA) is an essential step in data science that involves examining datasets to summarize their main characteristics using statistical methods and visual techniques. In this project, EDA is used to clean the data, handle missing values, and explore relationships between different variables. By using tools such as bar charts, boxplots, and violin plots, we can visualize patterns and trends in the dataset.

The main aim of this analysis is to identify key factors that affected the survival of passengers. For example, it is often believed that women and children were given priority during rescue operations. Similarly, passengers traveling in higher classes may have had better access to lifeboats compared to those in lower classes. Through systematic analysis, this project attempts to validate such assumptions using data.

Overall, this project demonstrates how data analysis techniques can be applied to real-world datasets to extract meaningful insights. It also highlights the importance of data preprocessing and visualization in understanding complex datasets and making informed conclusions.

❖ OBJECTIVES

1. To understand the structure and features of the Titanic dataset
2. To handle missing values and clean the dataset
3. To analyze survival rates based on gender, passenger class, and age groups
4. To visualize the data using graphs and charts
5. To identify key factors that influenced passenger survival

❖ METHODOLOGY

The methodology of this project follows a systematic approach to analyze the Titanic dataset using Exploratory Data Analysis (EDA) techniques. Initially, the dataset is collected and loaded into the Python environment using libraries such as Pandas and NumPy. In the data understanding phase, functions like head(), info(), and describe() are used to examine the structure of the dataset, identify data types, and obtain summary statistics. This helps in gaining a clear understanding of the data before performing any analysis.

The next step involves data cleaning, where missing values are handled to ensure data quality. The missing values in the Age column are filled using the median value, while missing values in the Embarked column are filled using the mode. The Cabin column is removed due to a large number of missing values, making it less useful for analysis. Additionally, column names are checked and corrected if necessary to avoid errors during processing.

After cleaning the data, feature engineering is performed by creating new variables to improve analysis. For example, the Age column is divided into different groups such as children, teenagers, adults, and senior citizens. This categorization helps in better understanding survival patterns across age groups.

In the exploratory data analysis phase, various relationships within the dataset are explored. Survival rates are analyzed based on gender, passenger class, and age groups using grouping and aggregation techniques. This step helps in identifying patterns and trends that influence survival chances.

Data visualization techniques are then applied to present the findings in a clear and understandable manner. Bar charts are used to compare survival rates, while boxplots and violin plots are used to analyze the distribution of age with respect to survival and passenger class. Heatmaps are also used to study correlations between numerical variables.

Finally, the results obtained from the analysis are interpreted to draw meaningful conclusions. The entire process helps in understanding the key factors that influenced passenger survival and demonstrates the importance of EDA in analyzing real-world datasets.

❖ Data Analysis

1. To understand the structure and features of the Titanic dataset

Code:

```
# 1. Import Libraries

import pandas as pd

import numpy as np

import seaborn as sns

import matplotlib.pyplot as plt

import warnings

warnings.filterwarnings('ignore')

# 2. Load Dataset

df=sns.load_dataset('titanic')

df

print("First 5 rows:")

print(df.head())

# 3. Basic Info

print("\nDataset Info:")

print(df.info())

print("\nStatistical Summary:")

print(df.describe())
```

Output:

First 5 rows:

```
   survived  pclass     sex   age  sibsp  parch     fare embarked  class  \
0         0       3    male  22.0      1      0   7.2500        S  Third
1         1       1  female  38.0      1      0  71.2833        C  First
2         1       3  female  26.0      0      0   7.9250        S  Third
3         1       1  female  35.0      1      0  53.1000        S  First
4         0       3    male  35.0      0      0   8.0500        S  Third

     who  adult_male deck  embark_town alive  alone
0    man        True  NaN  Southampton    no  False
1  woman       False    C    Cherbourg   yes  False
2  woman       False  NaN  Southampton   yes   True
3  woman       False    C  Southampton   yes  False
4    man        True  NaN  Southampton    no   True
```

Dataset Info:

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 15 columns):
#   Column       Non-Null Count  Dtype
---  ------       --------------  -----
0   survived     891 non-null    int64
1   pclass       891 non-null    int64
2   sex          891 non-null    object
3   age          714 non-null    float64
4   sibsp        891 non-null    int64
5   parch        891 non-null    int64
6   fare         891 non-null    float64
7   embarked     889 non-null    object
8   class        891 non-null    category
9   who          891 non-null    object
10  adult_male   891 non-null    bool
11  deck         203 non-null    category
12  embark_town  889 non-null    object
13  alive        891 non-null    object
14  alone        891 non-null    bool
dtypes: bool(2), category(2), float64(2), int64(4), object(5)
memory usage: 80.7+ KB
None
```

Statistical Summary:

	survived	pclass	age	sibsp	parch	fare
count	891.000000	891.000000	714.000000	891.000000	891.000000	891.000000
mean	0.383838	2.308642	29.699118	0.523008	0.381594	32.204208
std	0.486592	0.836071	14.526497	1.102743	0.806057	49.693429
min	0.000000	1.000000	0.420000	0.000000	0.000000	0.000000
25%	0.000000	2.000000	20.125000	0.000000	0.000000	7.910400
50%	0.000000	3.000000	28.000000	0.000000	0.000000	14.454200
75%	1.000000	3.000000	38.000000	1.000000	0.000000	31.000000
max	1.000000	3.000000	80.000000	8.000000	6.000000	512.329200

2. To handle missing values and clean the dataset

Code:

```
# 4. Missing Values

print("\nMissing Values:")

print(df.isnull().sum())

# 5. Data Cleaning

# Fill missing Age with median

df['age'] = df['age'].fillna(df['age'].median())

# Fill missing Embarked with mode

df['embarked'] =

df['embarked'].fillna(df['embarked'].mode()[0])

# Drop deck column (too many missing values)

df = df.drop(columns=['deck'])

# Verify again

print("\nMissing Values After Cleaning:")

print(df.isnull().sum())

# 6. Feature Engineering (Age Groups)

df['AgeGroup'] = pd.cut(df['age'],

                        bins=[0, 12, 18, 35, 60, 100],

                    labels=['Child','Teen','Young

Adult','Adult','Senior'])

print("\nAge Group Distribution:")

print(df['AgeGroup'].value_counts())
```

Output:

```
Missing Values:
survived        0
pclass          0
sex             0
age           177
sibsp           0
```

```
parch          0
fare           0
embarked       2
class          0
who            0
adult_male     0
deck         688
embark_town    2
alive          0
alone          0
dtype: int64

Missing Values After Cleaning:
survived       0
pclass         0
sex            0
age            0
sibsp          0
parch          0
fare           0
embarked       0
class          0
who            0
adult_male     0
embark_town    2
alive          0
alone          0
dtype: int64


Age Group Distribution:
AgeGroup
Young Adult    535
Adult          195
Teen            70
Child           69
Senior          22
Name: count, dtype: int64
```

3. To analyze survival rates based on gender, passenger class, and age groups

Code:

```
# 7. Survival Analysis

# Survival by Gender

print("\nSurvival Rate by Gender:")

print(df.groupby('sex')['survived'].mean())
```

```
# Survival by Class
print("\nSurvival Rate by Class:")
print(df.groupby('pclass')['survived'].mean())
# Survival by Age Group
print("\nSurvival Rate by Age Group:")
print(df.groupby('AgeGroup')['survived'].mean())
```

Output:

```
Survival Rate by Gender:
sex
female    0.742038
male      0.188908
Name: survived, dtype: float64

Survival Rate by Class:
pclass
1    0.629630
2    0.472826
3    0.242363
Name: survived, dtype: float64

Survival Rate by Age Group:
AgeGroup
Child          0.579710
Teen           0.428571
Young Adult    0.353271
Adult          0.400000
Senior         0.227273
Name: survived, dtype: float64
```

Interpretation:

Survival Rate by Gender : Females survived more than males.
Survival Rate by Class: 1st class had highest survival
Survival Rate by Age Group: Children survived more

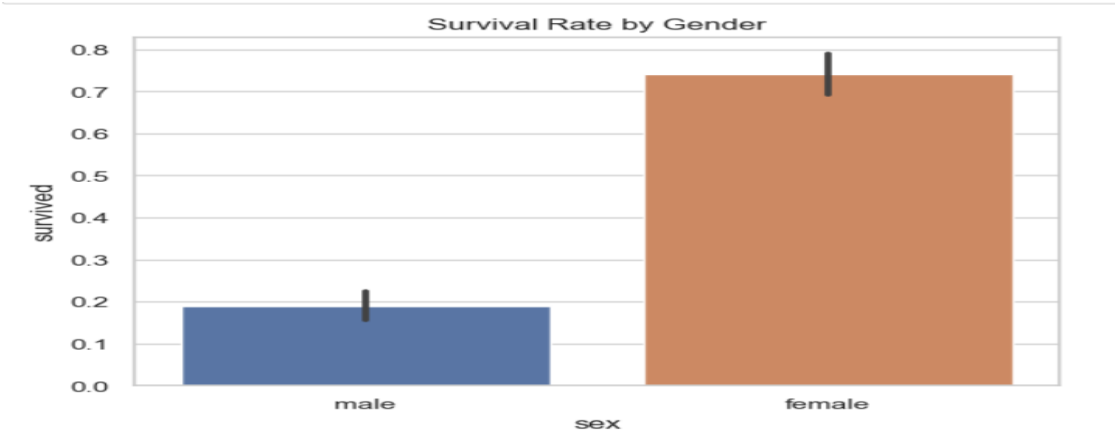
4. To visualize the data using graphs and charts

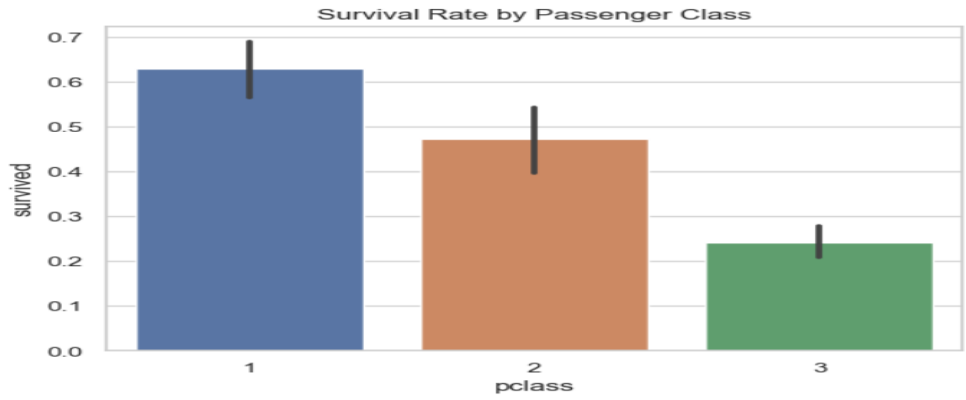
Code:

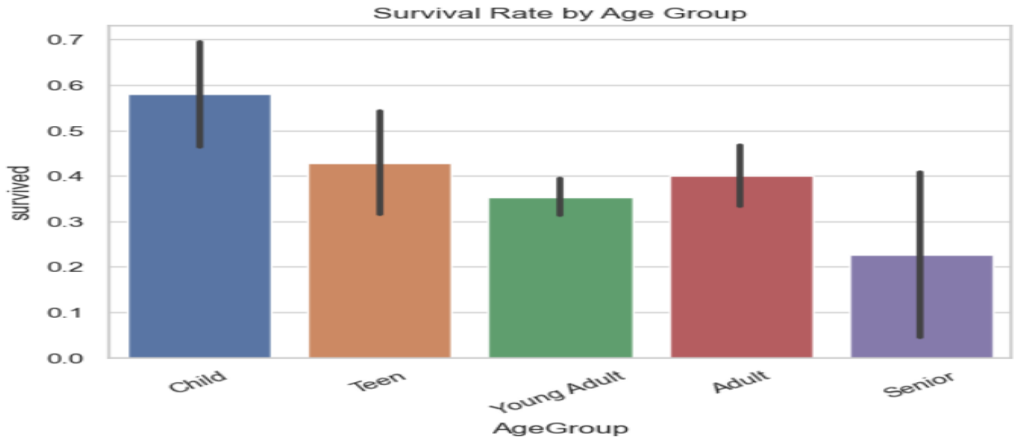
```
# 8. Visualizations
# Style
sns.set(style="whitegrid")

# 1. Survival by Gender
plt.figure()
sns.barplot(x='sex', y='survived', data=df)
plt.title("Survival Rate by Gender")
plt.show()
# 2. Survival by Class
plt.figure()
sns.barplot(x='pclass', y='survived', data=df)
plt.title("Survival Rate by Passenger Class")
plt.show()
# 3. Survival by Age Group
plt.figure()
sns.barplot(x='AgeGroup', y='survived', data=df)
plt.title("Survival Rate by Age Group")
plt.xticks(rotation=30)
plt.show()
# 5. Violin Plot - Age vs Class
plt.figure()
sns.violinplot(x='pclass', y='age', hue='survived', data=df, split=True)
plt.title("Age Distribution by Class and Survival")
plt.show()
```

Output:







Interpretation:

Survival by Gender : Women were much more likely to survive than men.This clearly shows a strong gender effect on survival.

Survival by Class: Higher-class passengers had better survival chances. This suggests rich or higher-class people were prioritized**.**

Survival by Age Group: Children survived the most. Old people (seniors) survived the least.This supports the idea of “women and children first”.

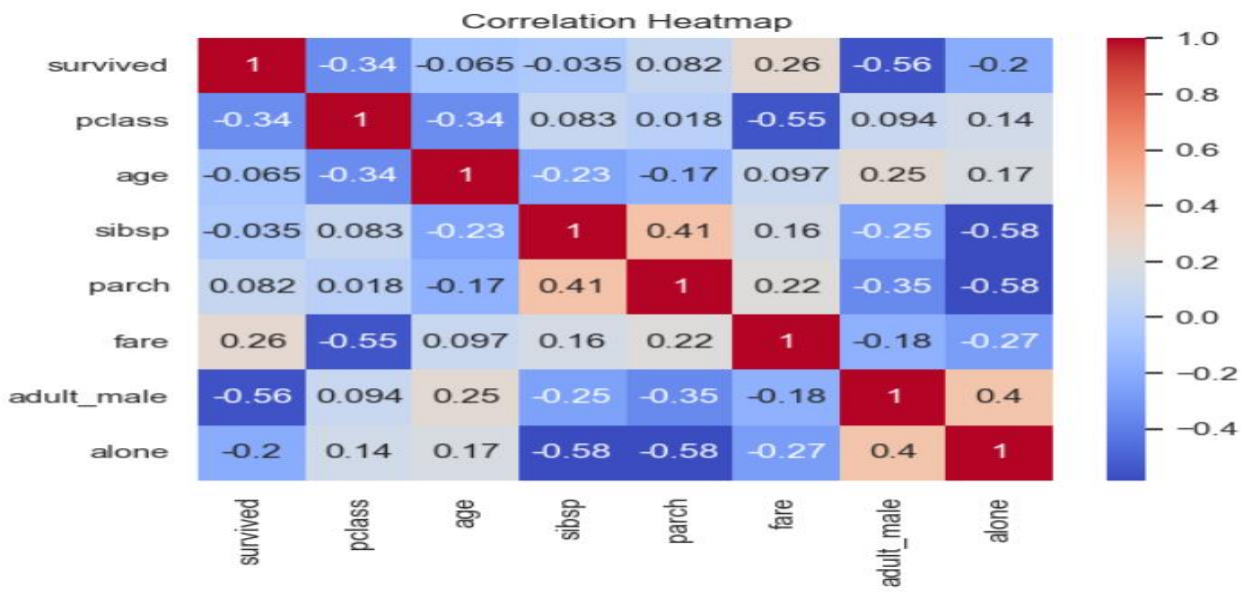
Violin Plot - Age vs Class: Lower class (3rd class) had more deaths. Higher class had better survival across all ages. Age matters, but class plays a bigger role.

5. To identify key factors that influenced passenger survival

Code:

```
# 9. Correlation Heatmap
plt.figure()
sns.heatmap(df.corr(numeric_only=True), annot=True, cmap='coolwarm')
plt.title("Correlation Heatmap")
plt.show()
```

Output:



Interpretation:

Heatmap showing relationships between variables. Pclass and Fare are related. Survival depends on multiple factors, not just one

❖ Overall Conclusion

The project “Titanic Survival Analysis using EDA” successfully demonstrates how data analysis techniques can be used to understand real-world datasets. The dataset was cleaned and analyzed to identify key factors influencing survival.

The analysis shows that gender, passenger class, and age significantly affected survival chances. Female passengers and first-class passengers had higher survival rates, while third-class passengers had lower chances of survival.

Data visualization helped in understanding patterns clearly and effectively. Overall, this project highlights the importance of EDA in extracting meaningful insights and supporting data-driven decisions.